

Given

$$V_{DD} = 5 \text{ V}$$

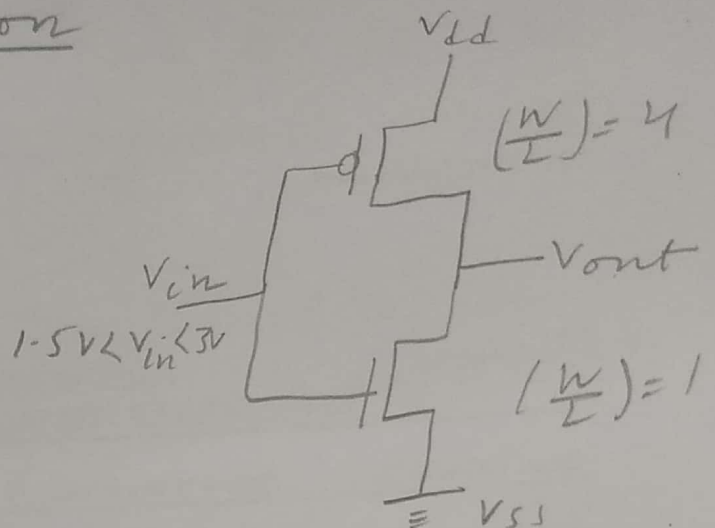
$$K_n' = 600 \mu\text{A/V}^2$$

$$K_p' = 240 \mu\text{A/V}^2$$

$$V_{tn} = 1 \text{ V} \quad \mu_n = 2.5 \mu_p$$

$$V_{tp} = -1 \text{ V}$$

Solution



a) $V_{OH} \rightarrow$ so PMOS in linear & NMOS in saturation

$$\frac{K_n'}{2} \left(\frac{W}{L} \right)_n (1.5 - 1)^2 = \frac{K_p'}{2} \left(\frac{W}{L} \right)_p \left[2(1.5 - 5 + 1)(V_{OH} - 5) - (V_{OH} - 5)^2 \right]$$

$$300 \times 1 \times (0.25) = 120 \times (4) \left[2(-2.5)(V_{OH} - 5) - (V_{OH} - 5)^2 \right]$$

$$\frac{75}{480} = \left[-5V_{OH} + 25 - V_{OH}^2 + 10V_{OH} \right]$$

$$0.156 = -V_{OH}^2 + 5V_{OH}$$

$$V_{OH}^2 - 5V_{OH} + 0.156 = 0$$

$$V_{OH} = 4.96 \text{ V}$$

$V_{OL} \rightarrow$ so PMOS in saturation & NMOS in linear

$$\frac{K_n'}{2} \left(\frac{W}{L} \right)_n \left[2(V_{in} - V_{tn})V_{OL} - V_{OL}^2 \right] = \frac{K_p'}{2} \left(\frac{W}{L} \right)_p (V_{in} - V_{DD} - V_{tp})^2$$

$$300 \times 1 \left[2(3 - 1)V_{OL} - V_{OL}^2 \right] = 120 \times 4 \left[3 - 5 + 1 \right]^2$$

$$4V_{OL} - V_{OL}^2 = \frac{48}{30} [1]$$

$$V_{OL}^2 - 4V_{OL} + 1.6 = 0$$

$$V_{OL} = 0.45 \text{ V}$$

$$\begin{aligned} \text{New } V_{OH} &= V_{OH} - 0.2 V_{DD} \\ &= 4.96 - 1 = \boxed{3.96} \text{ V} \end{aligned}$$

$$\begin{aligned} \text{New } V_{OL} &= V_{OL} + 0.05 V_{DD} \\ &= \boxed{0.79} \end{aligned}$$

Since $|NM_L| = |NM_H|$

$$V_M = \frac{3.96 + 0.79}{2} = \boxed{2.33} \text{ V}$$

$$\frac{k_n'}{2} \left(\frac{W}{L} \right)_n (2.33 - 1)^2 = \frac{k_p'}{2} \left(\frac{W}{L} \right)_p (2.33 - 5 + 1)^2$$

$$800 \left(\frac{W}{L} \right)_n (1.33)^2 = 240 \left(\frac{W}{L} \right)_p (-1.67)^2$$

$$\frac{\left(\frac{W}{L} \right)_n}{\left(\frac{W}{L} \right)_p} = \frac{240 \times 2.79}{600 \times 1.768} = \boxed{0.63}$$

$$\left[\frac{\left(\frac{W}{L} \right)_p}{\left(\frac{W}{L} \right)_n} = \frac{1}{0.63} = \boxed{1.587} \right]$$



± 2 volt noise has been added.

$V_{OH} = 3.96V$ $V_{OL} = 0.7V$ from part c

After corruption! for $V_{in} = 1.5V$ $V_{OH} = 3.96V$

$\rightarrow V_{OH}' = 3.96 - 2 = 1.96$ $V_M = 2.33V$

$\underbrace{\hspace{10em}}_{\text{less} < V_M}$

So At node ② signal will get corrupted

for $V_{in} = 3V$ $V_{OL} = 0.7V$

After corruption!

So $V_{OL}' = 0.7 + 2 = 2.7V$ $V_M = 2.33V$

$\underbrace{\hspace{10em}}_{\text{more} > V_M}$

Signal at node ② will get corrupted

8) When V_{DD} goes from 1.5 V to 3V, there is both static & dynamic power consumption.

$$\text{Static power} = V_{DD} \cdot I_{DP} \quad \text{PMOS in saturation}$$

$$I_{DP} = \frac{k_p}{2} \left(\frac{W}{L} \right)_p (3 - 1.5)^2$$

$$\underline{I_{DP}} = \frac{2.40}{2} (4) \times 1 = \underline{4.80 \text{ } \mu\text{A}}$$

$$\underline{P_{\text{static}} = 2.40 \text{ mW}}$$

$$\text{Dynamic power} = V_{DD}^2 \cdot C_L \cdot f \cdot \left(\frac{V_{OH} - V_{OL}}{\text{swing}} \right)$$

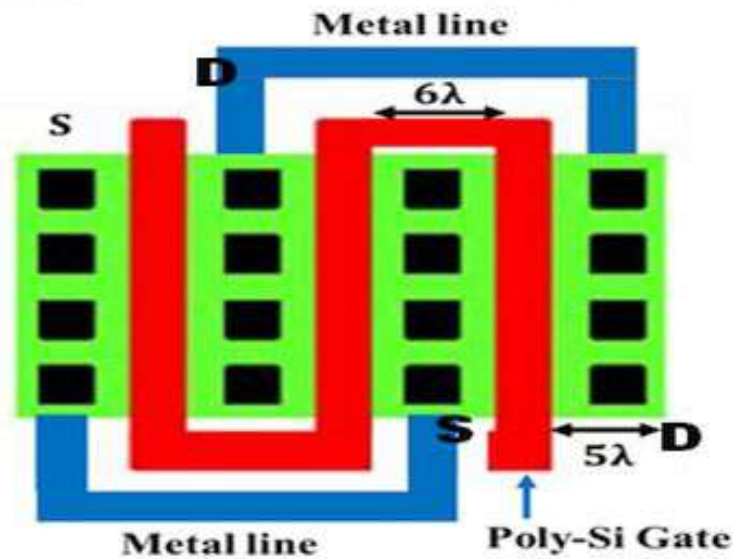
$$= 2.5 \times 10 \times 10^{-12} \times 10^7 (4 - 1.5 - 0.45)$$

$$= 2.5 \times 10^{-9} (4.51)$$

$$\underline{\text{Dynamic power}} = \underline{11.27 \text{ mW}}$$

Q1. A)

i.



ii) Perimeter drain = 62λ or $15.5 \mu\text{m}$
Area drain = $440\lambda^2$ or $27.5 \mu\text{m}^2$
 $C_{db} = \text{area} \times C_j + \text{Perimeter} \times C_{jsw}$
 $= 9.7 \text{ fF}$

iii) Photolithography.

Minimum device dimensions permit to use photolithography. e-beam & ion beam lithography are useful when few 10nm of nm .

II. Initially @ $\lambda = 250 \text{ nm}$, $V_{DD} = 2.2 \text{ V}$, $P = 5 \text{ W}$, $f_{\text{CLK}} = 100 \text{ MHz}$

a. Now $\lambda = 90 \text{ nm}$

$$\text{Scaling factor } (s) = \frac{250}{90} \approx 2.8$$

Constant voltage scaling so $V_{DD} = \underline{2.2 \text{ V}}$

$$P = V_{DD} I_{D\text{sat}} \quad (I_{D\text{sat}} \text{ increases by } s)$$

$$P = \cancel{250} \times 5 \text{ W} \approx \underline{13.9 \text{ W}}$$

① $\Rightarrow f_{\text{CLK}}$ is inversely proportional to intrinsic gate delay

$\swarrow$ $C_{\text{gate}} V_{DD} \rightarrow \text{constant}$
 $I_{D\text{sat}} \rightarrow \text{increases by } s$
 $\searrow$ scale down by s

f_{CLK} increases by s^2

$$\begin{aligned} \text{New } f_{\text{CLK}} &= s^2 \times 100 \text{ MHz} \\ &= \underline{\underline{784 \text{ MHz}}} \end{aligned}$$

b.

Physical dimension scaling (S) = 2.8

Voltage scaling (V) = $\frac{2.2}{1} = 2.2$

$$V_{DD} = \underline{1V}$$

$$P = V_{DD} I_{Dsat}$$

$$\downarrow \quad \quad \quad \downarrow$$
$$\frac{1}{V} \times \left(\frac{S}{V^2} \right) \rightarrow \frac{S}{V^3}$$

$$\mu_n \frac{\epsilon_{ox}}{t_{ox}} \cdot \frac{W}{L} (V_{in5} - V_{t0n})^2$$

$$P = \frac{2.8}{2.2^3} \times 5W \approx \underline{\underline{1.3W}}$$

$$f_{CLK} \propto \frac{I_{Dsat} \rightarrow \frac{S}{V^2}}{C_{gate} V_{DD} \rightarrow \frac{1}{V}} = \frac{S/V^2 \times V \times S}{\frac{1}{S}} = \frac{S^2}{V}$$

$$f_{CLK} = \frac{S^2}{V} \times 100 \text{ MHz} = 356 \text{ MHz}$$

c:

$P = \text{Constant}$

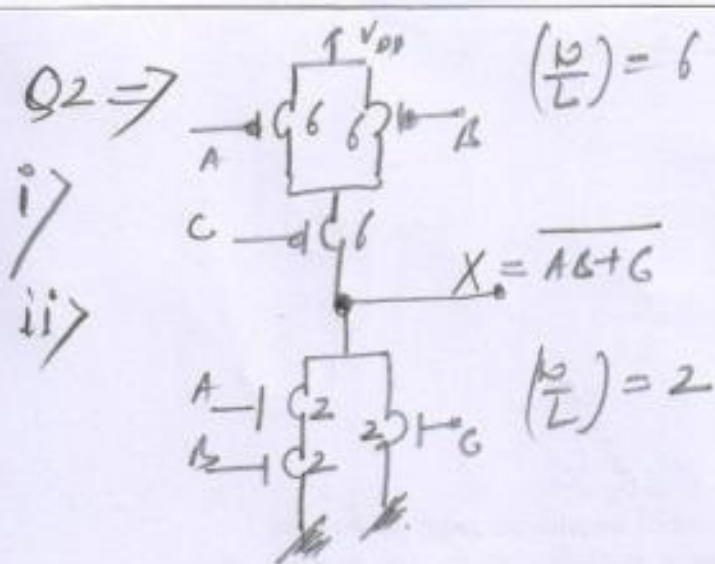
$$\text{i.e. } \frac{S}{V^3} = 1 \Rightarrow S = V^3$$

$$\Rightarrow 2.8 = V^3$$

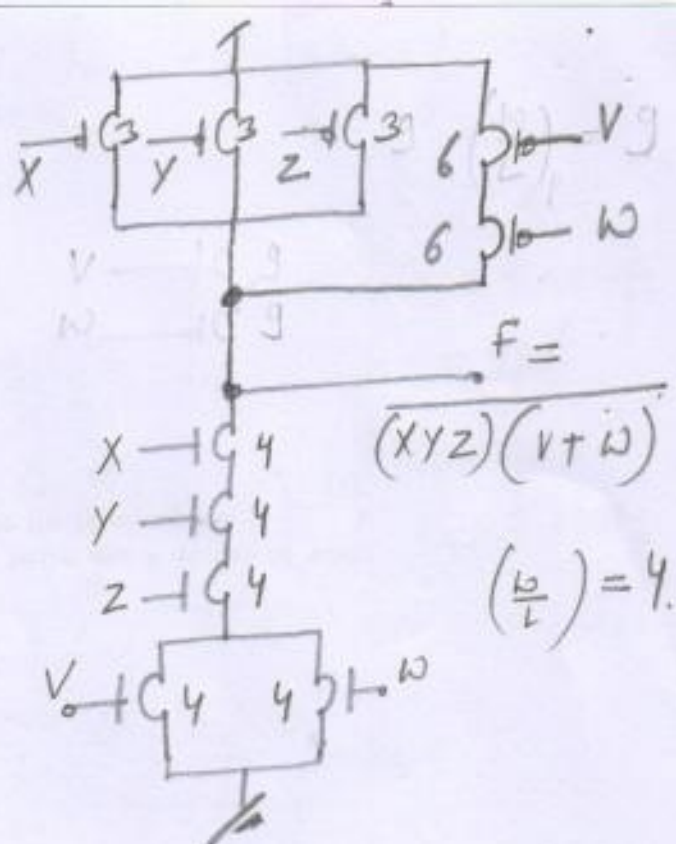
$$\Rightarrow V = 1.406$$

$$V_{DD} \approx \underline{\underline{1.6V}}$$

$$f_{CLK} = \frac{S^2}{V} \times 100 \text{ MHz} \approx \underline{\underline{558 \text{ MHz}}}$$



Note: $(\frac{W}{L})$ marked in schematic



iii) least delay path from A to F.

$$F = G.B.H = \left(\frac{8}{4}\right) \times \left(\frac{7}{4}\right) \times 1 \times 100 = 350$$

$$S = F^{1/A} \Rightarrow$$

$$\hat{N} = 4.58 \text{ or } 6$$

we need to insert $\frac{4}{6}$ inverters. otherwise logic will be inverted.

$$\hat{D} = \hat{N} \cdot F^{1/\hat{N}} + P$$

$$\hat{D} = 27.15 \text{ delay unit}$$

Also

$$\text{Absolute delay } \hat{D} = 13.57 \text{ ns}$$

$$\text{Max } f_{\text{req}} = \frac{1}{13.57 \text{ ns}} \approx 73 \text{ MHz}$$

$$iv) \text{ Dynamic power consumption} = P_{o-t} \cdot C_L \cdot V_{DD}^2 \cdot f$$

$$P_{o-t} = P_{o-t \text{ stage } 1} + P_{o-t \text{ stage } 2}$$

$$P_{o-t} = 0.23 + 0.084 = 0.318$$

$$P_{dyn} = 580.35 \times 10^{-6} \approx \underline{580 \mu W}$$

Given:

$$C_L = 400 \times 10^{-15} F \\ = 4 pF$$

$$V_{DD} = 2.5$$

$$f = 73 \text{ MHz}$$

$$v) s = \frac{250}{180} = 1.38, \quad V_{DD} = \frac{2.5}{1.38} = \underline{1.8}$$

$$C_{guin} = 7.25 pF, \\ C_L = 2.9 pF$$

$$\tau_{180nm} = \frac{0.6 ns}{1.38} = 0.362 ns$$

Given:

$$C_{guin} = 10 pF \\ \text{Cor. W.L.}$$

$$\tau' = \frac{\tau}{s}$$

$$f = \frac{1}{\tau} = \frac{1}{0.362} = 101 \text{ MHz}$$

$$\tau_{AB} = 9.82 ns, \quad f = \underline{101 \text{ MHz}}$$

$$P_{dyn} =$$

$$P_{dyn} = 301.7 \mu W$$